Momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate and compare signed momentum.

SUCCESS CRITERIA - I can...

1. Multiply mass by velocity with correct units.
2. Include direction or a sign.
3. Distinguish zero total momentum from no motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An asteroid's period is 5 years. Explain why that number alone cannot establish an Earth encounter. Name two additional observations needed.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $p_{total} = p_1 + p_2$.
Use signed velocities; momentum units are kg·m/s.

Worked reasoning: A 3 kg cart moving left at 2 m/s has momentum -6 kg·m/s when right is positive. The minus sign gives direction, not a negative mass.

Your turn: Draw separate velocity and momentum arrows, labeling their different units.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

The two carts have the same speed in opposite directions. Which momentum values are positive if right is positive?



2 kg; 3 m/s right

2 kg; 3 m/s left

Marzano check - circle: ① ② ③

04 Apply the physics

Use the two carts in question 3: each has mass 2 kg, and their velocities are +3 and -3 m/s. Complete the momentum table and find the total. Explain why a zero total does not mean either cart is stopped.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

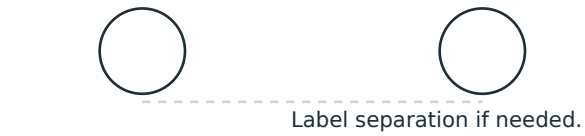
05 Explain your reasoning

One cart doubles mass at the same velocity. Another doubles speed at the same mass. Compare the momentum multipliers.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg test mass feels 24 N toward a source. Find g and the force on a 8 kg mass at the same point.
Label the masses, field or force relevant to the question.



Marzano check - circle: 1 2 3

07 Exit check

A 4 kg cart moves left at 2 m/s. State its momentum with direction.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Build two moving-cart examples whose total momentum is zero but whose masses differ. Draw both momentum arrows.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3